VITyarthi | Open Source Software

The Open Source Audit

A Capstone Project for OSS NGMC Course

Course: Open Source Software | Unit Coverage: 1 – 5
Max Marks: 100

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1. Introduction

Open source software is typically perceived as merely “free software,” but that phrase hardly touches the surface. Its actual usefulness extends well beyond saving money. At its foundation, open source is about openness, cooperation, and providing people actual power over the technology they use. Unlike proprietary software, which sometimes seems like a closed system where you can only utilize what you're given, open source welcomes you to explore inside, learn how things function, and even alter them to fit your requirements. It changes people from passive consumers into active players in the computer world.

This idea is prominently represented in the GNU General Public License (GPL), which is utilized by products like VLC Media Player. The GPL doesn’t simply make software accessible—it assures that it remains open, flexible, and shared for everyone. It safeguards the principle that once something is made free in terms of freedom, it should stay that way. In that sense, open source is more than simply a technique to produce software—it’s a strong statement about ownership, innovation, and making technology accessible to everyone, not just a chosen few.

Your choice	VLC Media Player _____

2. Project Report Structure

A1 — The Problem VLC Was Created to Solve (Expanded)

I never really thought about how media players operated or why certain files played while others didn't until I started this project. Like the majority of users, I thought a media player was simply a standard utility that was pre-installed on the computer. Actually, I used to believe that VLC Media Player was just one of those built-in default programs. I didn't aware VLC was really developed to address a very serious and annoying issue in digital media until I looked into its history.

Media playback was not as smooth in previous computer systems as it is now. Compatibility was a persistent problem, and various software was needed for different file types. When a user downloads a video file, they may discover that the necessary codec is missing, which prevents the movie from playing. Users had to install extra plugins or whole other media players in order to view their own material, which resulted in a confused experience. A large number of these media players were proprietary, which meant they were limited to operating systems or only supported certain formats. Users' freedom and control were thus restricted as a consequence of being indirectly compelled into certain ecosystems.

In a real-world setting like India, where users often deal with a range of file formats from various sources, this issue becomes much more pertinent. Not everyone has the technical know-how to handle codecs and setups or access to pricey software. A tool becomes very beneficial if it just "works" and doesn't need any further setup. By including a variety of codecs directly into the program, VLC specifically solved this problem. This made media playing simple for users of various backgrounds and did away with the requirement for external installs.

A group of students at École Centrale Paris were working on a project to broadcast video across a university network, which is where VLC had its start. The instruments available at the time were either too rigid or too constrained to meet their needs. The pupils decided to create their own solution rather than adjust to these limitations. This choice itself illustrates a key component of open-source thinking: when current tools are constrictive, developing and disseminating a better substitute is a logical reaction.

What began as a network streaming experiment eventually developed into a fully functional media player that can handle almost any format. VLC's main concept was to allow consumers total control over their material and eliminate reliance on proprietary technologies in addition to playing media. This is what makes VLC noteworthy: in addition to resolving a technical issue, it also addressed a more fundamental problem pertaining to software use freedom and accessibility.

Knowing this genesis tale, in my opinion, altered my outlook on software. I now see VLC as a solution that sprang from a real-world need and a philosophy that prioritizes transparency and user control rather than just a technology.

A2. The license — what it actually says

The well-known orange traffic cone of VLC Media Player sat quietly on my desktop for a long time. It was a dependable, free application that I hardly paid any care to. Although I was aware that it was "open source," like many others, I interpreted the word as nothing more than "free of cost." My perspective didn't entirely alter until I looked into the GNU General Public License (GPL), which is the license that governs it. I came to learn that open source is more than merely a pricing strategy. It is a philosophy that concerns ownership, freedom, and the interplay between people and technology.

The bulk of proprietary software works like a sealed black box. Users may connect with it, but they will never be able to grasp it. The developer has total control over the internal rationale, decisions, and methods. The GPL upends this paradigm by giving basic freedoms that transform the way people interact with software. It offers users the freedom to run the program for any purpose, investigate how it performs by examining its source code, adapt it to meet their own needs, and distribute both the original and customized versions to others. Users become active participants who may criticize, improve, and enhance the tools they rely on rather than being passive consumers. In this manner, open source extends beyond software and provides a basis for digital autonomy, giving individuals back control in a technological world that is getting more and more centralized.

The principle of copyleft is one of the GPL's most fundamental characteristics. Although it might at first sight look like a legal protection, it truly symbolizes a more deep ethical ideal based on reciprocity. Copyleft assures that people who change and distribute software under a GPL license must retain the same rights for others. This indicates that freedom cannot be controlled or monetized. Rather, it generates a continuing cycle in which every contribution builds upon the preceding one while still being accessible to users in the future. This paradigm fosters a dramatically different value system in a culture that is frequently dominated by ownership and exclusivity, one that places collaboration ahead of control and shared development ahead of personal advantage.

The statement "free as in free beer" versus "free as in freedom" sums up a popular misperception surrounding open source. Although VLC incorporates both, it is vital to differentiate between the two. Financial restrictions are erased when there is no expenditure, while intellectual and creative restraints are abolished when there is freedom. This more fundamental sort of freedom is reinforced by the GPL, which mandates that users be active in the creation of technology rather than merely being its clients.

The repercussions of this idea become considerably more relevant when examined from a larger aspect, especially in countries like India. Proprietary software may be unreasonably costly, which limits access to opportunities and information. These restrictions are overcome by open-source licenses like the GPL, which make powerful tools accessible to everyone. They make it feasible for students, independent developers, and small enterprises to connect with technology on an equal footing. By doing this, open source changes technology from a luxury into a communal platform for growth, innovation, and education.

In the end, the larger relevance of software like VLC derives from the GPL. It illustrates that software doesn't have to be a closed, static product under the control of a limited number of individuals. Alternatively, it may be a collaborative, dynamic system that is modified by its users. More crucially, it underscores that having access to tools is only one part of actual freedom in the digital age; another is the obligation to defend that access for others. Therefore, open source is more than merely a development paradigm; it is a committment to ensure that knowledge is available, shared, and continually evolving for future generations.

A3. The ethics of open source

The notion of open source extends beyond technical implementation and reaches the

realm of ethics. It raises crucial considerations about how information should be shared and who should have authority over technology. Before this project, I never really thought about these factors. I just used software based on convenience. However, investigating open source helped me understand that there is a greater philosophy behind it.

On one side, open source fosters transparency and cooperation. When the source code is open to everyone, it enables users to check what the program is doing, discover possible concerns, and suggest changes. This establishes a system where invention is not restricted to a single firm but is disseminated throughout a worldwide community. VLC is an excellent illustration of this, since its ability to handle multiple formats and systems is a consequence of contributions from developers across the globe.

However, the concept that all software should be open source is not totally practicable. Companies put enormous resources into research and development, and proprietary models give a mechanism to maintain these efforts. Without some type of funding, it would be impossible for many firms to continue inventing. This provides a balance between openness and corporate demands, where both models have their place.

Another ethical concern comes when firms construct lucrative products on top of open-source software. On one side, this might be considered as a good conclusion, since it highlights the usefulness of open-source foundations. On the other side, it raises issues about justice if firms gain from community donations without contributing anything back. From my opinion, the ethical option would be to give back to the community in some capacity, whether via code, financing, or help.

The expression “standing on the shoulders of giants” is especially pertinent in this scenario. Open source enables developers to build upon previous work rather than beginning from scratch, which speeds learning and creativity. since a student, I personally benefit from this system, since it offers me access to tools and information that would otherwise be impossible to attain. At the same time, it also develops a feeling of duty to respect and sustain the environment.

In conclusion, open source is not merely a development paradigm but a way of thinking about technology and cooperation. It fosters sharing, learning, and collaborative advancement, but also demanding ethical thought from those who use and profit from it. This project has made me more aware of these factors, and it has impacted the way I perceive the software I use every day.

Part B — Linux Footprint (Unit 2)

Before working on this subject, I had never really thought about how software is maintained or managed inside an operating system. I was more familiar with employing applications on Windows, where programs look as installed without disclosing what transpires behind the scenes. Exploring VLC Media Player on a Linux machine let me discover that Linux manages programs in a more systematic and transparent way.

On an Ubuntu-based system, VLC is installed using the APT package manager. Instead than downloading software manually, it may be installed directly from official repositories with a single command. This approach automatically handles dependencies, which makes installation simpler and more reliable. From my perspective, this seems more organized compared to manually installing software from multiple sources.

After installation, VLC is not preserved in a single location. Its components are scattered over multiple directories based on their function. The core executable is stored in /usr/bin, while supporting libraries are put in /usr/lib, and configuration files may be located in /etc or in the user directory ~/.config/vlc. Initially, this structure looked unusual, but it helped me discover how Linux distributes apps in a modular fashion.VLC functions under the existing logged-in user instead than requiring administrator credentials. It only starts when the user accesses it, either via the graphical interface or the terminal.

The update method is also done using the package manager. When the system is updated, VLC receives updates along with other programs. This centralized system assures that the software remains secure and up-to-date without necessitating human effort.
Overall, analyzing VLC in a Linux environment helped me appreciate how software is absorbed into the system. It showed me that programs are not merely solo tools, but part of a wider, well-organized structure that follows stated design principles.

Part C — The FOSS Ecosystem (Units 3 & 4)

Before this project, I thought of VLC media player as a standalone program. However, I found that it relies on various open-source software, primarily FFmpeg, which handles audio and video decoding. This taught me that even a basic media player is based on numerous layers of shared labor inside the open-source ecosystem.

VLC has also inspired other media players by creating a standard for supporting numerous formats without needing extra codecs. From my opinion, this made media playing more easier for those like me who before would not understand why certain files would not play. Although VLC is not a part of the LAMP stack, it is nonetheless linked to the wider online ecosystem through multimedia processing and streaming technologies.

The project is maintained by the VideoLAN community, where developers and users cooperate via forums and conversations. Unlike proprietary software, which is controlled by a single firm, VLC is created in a more open and collaborative fashion. Overall, this portion helped me grasp that open-source software is not generated separately, but as part of a linked system where many tools and communities contribute to each other.

Part D — Open Source vs Proprietary (Critical Analysis)

To better understand the practical value of VLC media player, it is useful to compare it with a proprietary alternative such as Windows Media Player. Both serve the purpose of media playback, but they differ significantly in terms of design philosophy, flexibility, and user control.

Dimension	Open Source	Proprietary Alternative
Cost	Completely free to use and distribute	Free but limited to Windows ecosystem
Security (who can audit the code?)	Source code is open for public auditing	Source code is closed and controlled
Support and reliability	Community-based support, forums and documentation	Official support from Microsoft
Freedom to modify	Users can modify and redistribute under GPL	No modification allowed
Control	Community-driven development	Controlled by a single company

4. Shell Script Tasks

Script 1 — System Identity Report (Unit 1 & 2)

Starter structure

```
  GNU nano 7.2                                          script1.sh
#!/bin/bash
STUDENT_NAME="Krishna Sadhwani"
SOFTWARE_CHOICE="VLC Media Player"

KERNEL=$(uname -r)
USER_NAME=$(whoami)
HOME_DIR=$HOME
UPTIME=$(uptime -p)
DATE_TIME=$(date)
DISTRO=$(lsb_release -d 2>/dev/null | cut -f2)

echo "========================================"
echo " Open Source Audit - $STUDENT_NAME"
echo "========================================"
echo "Chosen Software : $SOFTWARE_CHOICE"
echo "Distribution    : $DISTRO"
echo "Kernel Version  : $KERNEL"
echo "User            : $USER_NAME"
echo "Home Directory  : $HOME_DIR"
echo "Uptime          : $UPTIME"
echo "Date & Time     : $DATE_TIME"
echo "========================================"
```

```
krishna_sadhwani@Krishna-Sadhwani:~$ cd ~
krishna_sadhwani@Krishna-Sadhwani:~$ nano script1.sh
krishna_sadhwani@Krishna-Sadhwani:~$ chmod +x script1.sh
krishna_sadhwani@Krishna-Sadhwani:~$ ./script1.sh
========================================
 Open Source Audit - Krishna Sadhwani
========================================
Chosen Software : VLC Media Player
Distribution    : Ubuntu 24.04.4 LTS
Kernel Version  : 6.6.87.2-microsoft-standard-WSL2
User            : krishna_sadhwani
Home Directory  : /home/krishna_sadhwani
Uptime          : up 6 minutes
Date & Time     : Thu Mar 26 11:26:39 UTC 2026
========================================
krishna_sadhwani@Krishna-Sadhwani:~$ nano script1.sh
krishna_sadhwani@Krishna-Sadhwani:~$ nano script1.sh
krishna_sadhwani@Krishna-Sadhwani:~$
```

Script 2 — FOSS Package Inspector (Unit 2)

```bash
#!/bin/bash
# Script 2: FOSS Package Inspector
# Author: Krishna Sadhwani

# --- Package name ---
PACKAGE="vlc"

# --- Check if package is installed ---
if dpkg -l | grep -qw $PACKAGE; then
    echo "$PACKAGE is installed."

    # Show version, license, summary
    dpkg -s $PACKAGE | grep -E "Version|Maintainer|Description"
else
    echo "$PACKAGE is NOT installed."
fi

echo ""

# --- Case statement (philosophy note) ---
case $PACKAGE in
    vlc)
        echo "VLC: a free media player that removes format restrictions and promotes accessibility."
        ;;
    firefox)
        echo "Firefox: supports an open and privacy-focused web."
        ;;
    apache2)
        echo "Apache: powers a large part of the open internet."
        ;;
    mysql-server)
        echo "MySQL: an open-source database used worldwide."
        ;;
    *)
        echo "Unknown package."
        ;;
esac
```

```
krishna_sadhwani@Krishna-Sadhwani:~$ ./script2.sh
vlc is installed.
Maintainer: Ubuntu Developers <ubuntu-devel-discuss@lists.ubuntu.com>
Version: 3.0.20-3build6
Description: multimedia player and streamer
Original-Maintainer: Debian Multimedia Maintainers <debian-multimedia@lists.debian.org>

VLC: a free media player that removes format restrictions and promotes accessibility.
```

This script verifies whether the system has a given open-source package installed, in this case VLC. It looks for the package using the dpkg command and an if-else condition. Important data like the version and maintainer information are displayed if the package is found. A case statement that illustrates the package's purpose in the open-source ecosystem is also included in the script. This script explains how to leverage pipes, pattern matching, conditional statements, and command-line tools in shell scripting.

Script 3 — Disk and Permission Auditor (Unit 2)

```
# Script 3: Disk and Permission Auditor
# Author: Krishna Sadhwani

# --- List of important directories ---
DIRS=("/etc" "/var/log" "/home" "/usr/bin" "/tmp")

echo "========================================"
echo "        Directory Audit Report"
echo "========================================"

# --- Loop through directories ---
for DIR in "${DIRS[@]}"; do
    if [ -d "$DIR" ]; then
        # Get permissions, owner, group
        PERMS=$(ls -ld $DIR | awk '{print $1, $3, $4}')

        # Get size
        SIZE=$(du -sh $DIR 2>/dev/null | cut -f1)

        echo "$DIR => Permissions: $PERMS | Size: $SIZE"
    else
        echo "$DIR does not exist"
    fi
done

echo ""
echo "========================================"
echo "Checking VLC config directory..."
echo "========================================"

# --- Check VLC config directory ---
if [ -d "$HOME/.config/vlc" ]; then
    ls -ld "$HOME/.config/vlc"
else
    echo "VLC config directory not found."
fi
```

```
krishna_sadhwani@Krishna-Sadhwani:~$ chmod +x script3.sh
krishna_sadhwani@Krishna-Sadhwani:~$ ./script3.sh
========================================
        Directory Audit Report
========================================
/etc => Permissions: drwxr-xr-x root root | Size: 4.4M
/var/log => Permissions: drwxrwxr-x root syslog | Size: 57M
/home => Permissions: drwxr-xr-x root root | Size: 52K
/usr/bin => Permissions: drwxr-xr-x root root | Size: 98M
/tmp => Permissions: drwxrwxrwt root root | Size: 28K

========================================
Checking VLC config directory...
========================================
VLC config directory not found.
```

This script analyzes essential system folders by evaluating their permissions and disk use. It utilizes a for loop to run over a list of directories and confirms their presence using an if condition. The ls -ld program is used combined with awk to extract permission and ownership data, while the du function estimates the disk use. The script also checks for the VLC configuration directory in the user's home folder. This shows the usage of loops, conditional expressions, and command-line tools in shell scripting.ng.

Script 4 — Log File Analyzer (Units 2 & 5)

```
  GNU nano 7.2                                                    script4.sh *
#!/bin/bash
# Script 4: Log File Analyzer
# Author: Krishna Sadhwani

LOGFILE=$1
KEYWORD=${2:-"error"}
COUNT=0

# --- Check if file exists ---
if [ ! -f "$LOGFILE" ]; then
    echo "Error: File not found."
    exit 1
fi

# --- Read file line by line ---
while IFS= read -r LINE; do
    if echo "$LINE" | grep -iq "$KEYWORD"; then
        COUNT=$((COUNT + 1))
    fi
done < "$LOGFILE"

# --- Output ---
echo "-----------------------------------"
echo "Keyword '$KEYWORD' found $COUNT times"
echo "-----------------------------------"

echo ""
echo "Last 5 matching lines:"
grep -i "$KEYWORD" "$LOGFILE" | tail -n 5
```

```
krishna_sadhwani@Krishna-Sadhwani:~$ ./script4.sh /var/log/syslog error
-----------------------------------
Keyword 'error' found 17 times
-----------------------------------

Last 5 matching lines:
2026-03-26T11:44:45.315473+00:00 Krishna-Sadhwani kernel: WSL (1 - init(docker-desktop)) ERROR: ConfigApplyWindowsLibPath:2119: open /etc/ld.so.conf.d/ld.ws
l.conf failed 2
2026-03-26T11:46:16.426251+00:00 Krishna-Sadhwani kernel: WSL (1 - init(docker-desktop)) ERROR: ConfigApplyWindowsLibPath:2119: open /etc/ld.so.conf.d/ld.ws
l.conf failed 2
2026-03-26T11:47:24.711527+00:00 Krishna-Sadhwani kernel: WSL (1 - init(docker-desktop)) ERROR: ConfigApplyWindowsLibPath:2119: open /etc/ld.so.conf.d/ld.ws
l.conf failed 2
2026-03-26T11:49:20.255830+00:00 Krishna-Sadhwani systemd[1]: apport-autoreport.path - Process error reports when automatic reporting is enabled (file watch
) was skipped because of an unmet condition check (ConditionPathExists=/var/lib/apport/autoreport).
2026-03-26T11:49:20.255836+00:00 Krishna-Sadhwani systemd[1]: apport-autoreport.timer - Process error reports when automatic reporting is enabled (timer bas
ed) was skipped because of an unmet condition check (ConditionPathExists=/var/lib/apport/autoreport).
krishna_sadhwani@Krishna-Sadhwani:~$
```

This script scans a system log file to count how many times a given term occurs. It takes the log file as a command-line parameter and utilizes a while loop to read the file line by line. An if condition with the grep command is used to look for matches in a case-insensitive way. A counter variable is used to monitor the number of occurrences. Additionally, the script publishes the latest five matched lines from the log file. This shows the usage of loops, conditional expressions, command-line parameters, and text processing in shell scripting.

Script 5 — The Open Source Manifesto Generator (Unit 5)

```
  GNU nano 7.2                                                    script5.sh
#!/bin/bash
# Script 5: Open Source Manifesto Generator
# Author: Krishna Sadhwani

echo "========================================"
echo " Open Source Manifesto Generator"
echo "========================================"
echo ""

# --- Take user input ---
read -p "1. Name one open-source tool you use every day: " TOOL
read -p "2. In one word, what does 'freedom' mean to you? " FREEDOM
read -p "3. Name one thing you would build and share freely: " BUILD

# --- Date and output file ---
DATE=$(date '+%d %B %Y')
OUTPUT="manifesto_$(whoami).txt"

# --- Generate manifesto ---
echo "----------------------------------------" > $OUTPUT
echo "Open Source Manifesto - $DATE" >> $OUTPUT
echo "----------------------------------------" >> $OUTPUT
echo "" >> $OUTPUT

echo "I use $TOOL regularly, and it reflects the power of open-source collaboration in my daily life." >> $OUTPUT
echo "For me, freedom means $FREEDOM, especially in the context of technology and knowledge sharing." >> $OUTPUT
echo "If given a chance, I would build $BUILD and share it openly so others can learn, modify, and improve it." >> $OUTPUT
echo "Open source is not just about code - it represents transparency, accessibility, and collective progress." >> $OUTPUT

# --- Show result ---
echo ""
echo "Manifesto saved to $OUTPUT"
echo ""
cat $OUTPUT
```

```
krishna_sadhwani@Krishna-Sadhwani:~$ nano script5.sh
krishna_sadhwani@Krishna-Sadhwani:~$ chmod +x script5.sh
./script5.sh
========================================
 Open Source Manifesto Generator
========================================

1. Name one open-source tool you use every day: VLC
2. In one word, what does 'freedom' mean to you? control
3. Name one thing you would build and share freely: AI-based coding platform

Manifesto saved to manifesto_krishna_sadhwani.txt

----------------------------------------
Open Source Manifesto - 26 March 2026
----------------------------------------

I use VLC regularly, and it reflects the power of open-source collaboration in my daily life.
For me, freedom means control, especially in the context of technology and knowledge sharing.
If given a chance, I would build AI-based coding platform and share it openly so others can learn, modify, and improve it.
Open source is not just about code - it represents transparency, accessibility, and collective progress.
```

3. Project Timeline

This timeline is a guide, not a rigid rule. Some students will move faster on the technical parts; others will spend more time on the philosophical sections. Adjust based on your strengths — but do not leave the scripts to the final stretch.

Day	Focus	Tasks	Deliverable
1	**Setup**	Selected VLC Media Player as the project software. Researched its origin story and read about its licensing. Set up Linux environment using WSL and installed the software.	*Choice confirmed*
2-3	**Philosophy**	Wrote Part A1 focusing on the origin story of VLC and the problem it was created to solve. Reflected on real-world context and user needs.	*Draft A1*
3	**License & Ethics**	Completed Part A2 by analyzing the license and understanding software freedoms. Wrote Part A3 discussing ethical aspects of open-source with personal perspective.	*Draft A2 + A3*
4	**Linux Footprint**	Installed and explored VLC in Linux. Documented directories, permissions, and system behavior. Prepared screenshots and wrote Part B.	*Draft Part B + screenshots*
5	**Scripts 1 , 2 &3**	Developed and tested Script 1 (System Identity Report), Script 2 (Package Inspector), and Script 3 (Disk & Permission Auditor). Added explanations in report.	*Scripts 1-3 working*
6	**Scripts 4 & Ecosystem**	Implemented Script 4 (Log File Analyzer) and tested using real logs. Also researched VLC's dependencies, community, and ecosystem for Part C.	*Scripts 4 working +and draft part C*
7	**Comparison & Script 5**	Completed Part D by comparing VLC with proprietary alternatives. Developed Script 5 (Manifesto Generator).	*Draft Part D + Script 5*
8	**Report polish**	Compiled full report. Added introduction, conclusion, and screenshots. Ensured clarity, originality, and proper formatting.	*Complete draft*
9	**Submission**	Performed final review. Submitted report PDF, GitHub repository, and scripts. Prepared to explain scripts if required.	*Final submission*

4. Reference Resources

VideoLAN Organization - VLC Media Player Official Website -
https://www.videolan.org/vlc/

VLC Documentation –
https://vlc-user-documentation.readthedocs.io/

Report on VLC Media Player -
https://www.scribd.com/document/199514142/Report-on-Vlc

YouTube Video –
https://www.youtube.com/watch?v=Ky45V6b3aDo

GNU Project - Free Software Definition –
https://www.gnu.org/philosophy/free-sw.html

Eric S. Raymond - Cathedral and the Bazaar –
http://catb.org/~esr/writings/cathedral-bazaar

SPDX License List –
https://spdx.org/licenses

A Closing Thought

When I began this project, I didn't really think much about where the things I use every day originate from. Things like code editors, browsers, or even media players like VLC always seemed like they were simply "there." But while working on this audit, I learned that practically everything I use is based on open-source principles.

These tools exist because someone, somewhere, chose not merely to make something valuable, but to share it publicly so others may learn from it, enhance it, and use it freely. That notion fundamentally impacted how I look at software. It's not just about coding anymore – it's about teamwork, trust, and constructing something greater than yourself.

Understanding this helps me feel like I'm not only learning how to utilize technology, but also becoming part of a bigger community that believes in sharing information. And honestly, it seems like one of the most significant things I've taken away from this endeavor.